The Organizing Theme Andrews–Askey–Roy Does Not Name

A research programme: the Weyl algebra as the spine of the special-function pile, and containers as its categorification

Abstract

A standard special-functions reference reads as a jumbled pile because it presents the union of four mutually incompatible organizing themes and commits to none: the Askey scheme (order by parameter degeneration), monodromy (order by singularities on $\mathbb{P}^1$), Lie theory (order by which group's representations), and the holonomic/D-module view (order by rank). This note argues for the fourth, on the ground that it is the only one of which we already possess a categorification. Its central relation, $[\partial, x] = 1$, is exactly $DU - UD = I$ on Young's lattice — Stanley's differential-poset axiom, the corollary targeted by `lascoux/lean-plan.md` step 1.8, and the identity that Fomin's local rules prove bijectively. The q-layer closes the loop with the rest of `q-bool/`: the q-Weyl relation $D_q M_x - q\, M_x D_q = 1$ is the Jackson row already present in the decategorification table of `q-container.tex`. The programme is therefore to climb the Askey scheme from its bottom — Hermite, the Fock representation, $\mathcal{A}_1$ — towards its top — Askey–Wilson, Zhedanov's AW(3) — carrying a container model up with us, and to stop honestly wherever the carrying fails. Five phases, each with a deliverable and a falsification condition. Phase C is flagged in advance as the likely point of failure, and §6 says why.

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1 The complaint, and the diagnosis

The complaint is that a comprehensive special-functions reference — Andrews, Askey and Roy [1] is the one at hand — presents a pile. Chapter after chapter of families, each with its own orthogonality, recurrence, integral representation and generating function, and no evident reason why these families and not others.

The diagnosis is not that the book is disorganised. It is that *four* organizing themes exist, they do not agree, and a comprehensive book presents their union.

theme	orders the pile by	the chart	canonical source
Askey scheme	degeneration of parameters	a poset of limits	[2]
monodromy	singularities on $\mathbb{P}^1$	confluence of punctures	[8]
Lie theory	which group's irreps	group contraction	[3, 7]
holonomic / D-module	rank over the Weyl algebra	*a finiteness theorem*	[9, 4]

They genuinely disagree. The Askey scheme orders by what happens as a parameter tends to a limit; monodromy orders by how many punctures the differential equation has; Lie theory orders by the dimension and type of a symmetry group; holonomy orders by a rank, and produces not a chart but a decision procedure. Any two of these cut the pile differently, which is why no two books' tables of contents agree.

Remark 1.1 (the cheap fix, stated first). If the immediate want is relief rather than a programme: the Askey scheme is a literal one-page directed graph, and Koekoek–Lesky–Swarttouw [2] is organized *as* the chart rather than around it, with the q-scheme as a parallel chart that [1] does not have. Thirteen classical families become one poset with Askey–Wilson on top and Hermite at the bottom. Half an hour, and the pile is a poset. That is worth doing before anything below.

2 The thesis

Thesis 2.1 (*ours*). Of the four themes, the holonomic one is the organizing theme to adopt, because it is the only one whose central object we already possess a categorification of. Its central object is the Weyl algebra; our categorification of the Weyl algebra is Fomin's local rule, read as a vertical container isomorphism.

The remainder of §2 establishes the three links this depends on. None of the three is new; what is new, as far as we can tell, is putting them in one line.

2.1 Link 1: $DU - UD = I$ is the Weyl algebra

Definition 2.2. The (first) Weyl algebra is $\mathcal{A}_1 = \mathbb{C}\langle x, \partial\rangle$ subject to $\partial x - x\partial = 1$.

Proposition 2.3 (standard)**.** *Let $\Lambda = \mathbb{C}[p_1, p_2, \dots]$ be the ring of symmetric functions and identify $\mathbb{C}[\mathit{Young}] \cong \Lambda$ by $\lambda \mapsto s_\lambda$. Then*

$$U = \textit{multiplication by } p_1, \qquad D = \partial/\partial p_1$$

realise the up and down operators of Young's lattice — $Us_\lambda = \sum_{\mu \gtrdot \lambda} s_\mu$ by the Pieri rule, and D is its adjoint for the Hall inner product — and satisfy $DU - UD = I$. This is Stanley's differential-poset axiom [5].

Remark 2.4 (a multiplicity space, and a question). Be precise about what representation this is. As an $\mathcal{A}_1$-module, $\Lambda = \mathbb{C}[p_1] \otimes \mathbb{C}[p_2, p_3, \dots]$, with $\mathcal{A}_1$ acting on the first factor only. So Young's lattice carries the Fock representation *with multiplicity space* $\mathbb{C}[p_2, p_3, \dots]$; the higher power sums are spectators. The polynomial ring $\mathbb{C}[x]$ with $x, d/dx$ — whose orthogonal basis for the Gaussian measure is the Hermite polynomials, the bottom of the Askey scheme — is the same representation with trivial multiplicity. Hence Question 8.1 below.

2.2 Link 2: Fomin's rule is the bijective proof of that relation

Counting the fibres of the vertical container isomorphism of `hypergeometric.tex` §5.2 gives

$$|U\alpha \cap U\beta| = \delta_{\alpha\beta} + |D\alpha \cap D\beta|,$$

which is $DU - UD = I$ coefficientwise. The $\delta_{\alpha\beta}$ is the growth-diagram label $x = 1$ being legal at exactly one face shape. This is step 1.8 of `lascoux/lean-plan.md`, and steps 1.3–1.5 are the rule, its reverse, and the two round trips; 1.3 is complete.

Remark 2.5. The content of Link 2 is that the relation of Definition 2.2 has a *bijective* proof, uniform in (α, β), in which the two sides are fibres of one container and the relation is an isomorphism between them rather than an equation between numbers. That is what makes Thesis 2.1 more than a restatement: we do not merely know the relation, we have a morphism that witnesses it.

2.3 Link 3: holonomic means finite rank over the Weyl algebra

Proposition 2.6 (reported, [9])**.** *A function is holonomic when it is annihilated by a left ideal of the Weyl algebra $\mathcal{A}_n$ of maximal possible dimension — equivalently, when the $\mathcal{A}_n$-module it generates has finite holonomic rank. Holonomic functions are closed under sum, product, and definite summation or integration over a variable; identities between them are decidable, and the Wilf–Zeilberger machinery is the decision procedure.*

So this theme does not produce a chart. It produces a finiteness statement: the pile is not a pile but the set of finite-rank modules over one algebra, and the apparent profusion of families is the profusion of ideals, not of phenomena. The reason for preferring it here is Thesis 2.1: *that algebra* is the one we can categorify.

2.4 The square

$$\begin{array}{ccccc}
\textbf{Cont}\ (\text{vertical isos}) & \xrightarrow{\text{count fibres}} & \mathcal{A}_1\text{-modules} & \xrightarrow{\text{generating function}} & \text{special functions} \\
\Big\downarrow{\scriptstyle q\text{-deform}} & & \Big\downarrow{\scriptstyle q\text{-deform}} & & \Big\downarrow{\scriptstyle q\text{-deform}} \\
\textbf{Cont}_q & \xrightarrow{\text{count fibres}} & \mathcal{A}_1^q\text{-modules} & \longrightarrow & q\text{-special functions}
\end{array}$$

The top-left corner is `lascoux/`; the top-middle is Stanley and Zeilberger; the top-right is [1]. The bottom row is `q-bool/`. Every arrow in the top row exists. The claim of the programme is that the rest of the square can be filled in, and the phases of §4 are the attempts.

3 The q-layer is already half built

Proposition 3.1. *Let $D_q f(x) = \dfrac{f(qx) - f(x)}{(q-1)x}$ be the Jackson derivative and M_x multiplication by x. Then*

$$D_q M_x - q\, M_x D_q = 1.$$

Proof. $D_q(xf) = \dfrac{qxf(qx) - xf(x)}{(q-1)x} = \dfrac{qf(qx) - f(x)}{q-1}$ and $xD_qf = \dfrac{f(qx) - f(x)}{q-1}$, so $D_q(xf) - q\,xD_qf = \dfrac{qf(qx) - f(x) - qf(qx) + qf(x)}{q-1} = f(x)$. **verified** □

Corollary 3.2 (*ours*)**.** *The decategorification table of* `q-container.tex` *contains the row "derivative ∂ (hyperplanes) $\leftrightarrow$ Jackson D_q" and the row "$\mathrm{Ind}_*, \mathrm{Res}_* \leftrightarrow$ multiplication, comultiplication". Together these are the two generators of Proposition 3.1. So a container model of the q-Weyl algebra is not a new construction to be invented: its generators are already in the table, and what is missing is the relation — the q-analogue of Link 2.*

This is the single most encouraging fact in this note, and it is what makes Phase D cheap rather than speculative.

3.1 The top of the scheme

Remark 3.3 (reported). The Askey scheme has a top as well as a bottom. Askey–Wilson polynomials sit above every other family in [2], and the algebra governing them is Zhedanov's Askey–Wilson algebra AW(3) [10, 6] — three generators with q-commutator relations, degenerating down the scheme as the parameters degenerate.

So the programme has a shape: $\mathcal{A}_1$ at the bottom (Hermite, where we have a categorification), AW(3) at the top (Askey–Wilson, where we have none), and the Askey scheme as the path between. Climb it carrying the container model, and record where the model stops being carried.

4 The programme

Five phases. Each states its deliverable, its dependencies, an honest confidence that it will work, and — the part that matters — what observation would kill it.

Phase 0 — orientation (days)

- **0.1** Read the chart in [2], per Remark 1.1. Deliverable: the poset, drawn, with the q-scheme beside it.
- **0.2** Read [9]. Twenty pages, and it is the thesis statement for the chosen theme.
- **0.3** Settle Question 3.9 of `hypergeometric.tex`: is the class of q-containers closed under the Hadamard product, pulled back along dim? Cheap, independent of everything else, and needed by Phase D.

Confidence: n/a, this is reading. **Cost:** under a week.

Phase A — the Weyl algebra at the container level

Deliverable. Finish `lascoux/` Phase 1 (steps 1.4–1.8): the reverse local rule, both round trips, the assembly as a vertical morphism in **Cont**, and $DU - UD = I$ as a corollary with no `sorry`. Then the harder half: a container model of $\mathcal{A}_1$ *as an algebra*, not merely of its defining relation — objects for x and ∂, a composition, and the relation as a specified isomorphism.

Dependencies. None. Step 1.3 is complete.

Confidence. Steps 1.4–1.8: **high** (85%). The reverse rule is the forward rule's six cases run backwards, and the only case needing new work is the bounce, where `diffRow` needs a counterpart that finds the row to *remove*. The algebra model: **medium** (50%), because composition of vertical morphisms over different shape pairs is exactly the composability problem that thesis §4.2 records as the optic obstruction.

Falsified if the round trips of step 1.5 fail — i.e. if the rule as stated is not a bijection on the nose but only up to something. That would not be a disaster (it would make the structure a lens rather than an iso, which §4.2 half expects) but it would change every statement downstream from "isomorphism" to "optic".

Phase B — WZ certificates as container morphisms

Deliverable. An answer to Question 5.10 of `hypergeometric.tex`: is a Wilf–Zeilberger certificate a vertical morphism in **Cont** over the face data of a two-dimensional growth diagram? Calibration case: q-Vandermonde, laid out as a growth diagram rather than as a recurrence, whose bijection is already known (the subspace-intersection argument, `hypergeometric.tex` Cor. 6.6). The test is sharp: *does the WZ certificate for q-Vandermonde reproduce the subspace bijection?*

Dependencies. Phase A for the notion of vertical morphism; Phase 0.2.

Confidence. Medium-low (35%) in general, **high** (75%) for the calibration case alone. The general obstruction is that WZ certificates are rational functions and need not take integer values, so there need be no bijection to find. The reason to try anyway is the payoff: if it works, a decision procedure becomes a machine for producing container isomorphisms.

Falsified if the q-Vandermonde certificate is integral but demonstrably not the subspace bijection, *and* no reindexing relates them. Then WZ and Fomin are cousins and §5 of `hypergeometric.tex` should end at its Warning 5.7.

Phase C — the Askey scheme as a poset of containers

Deliverable. A container-level meaning for the *edges* of the Askey scheme. Each edge is a parameter limit; the question is what a limit of containers is.

Dependencies. Phases A and B; Phase 0.1.

Confidence. Low (15%), and this is flagged in advance as the likely point of failure of the whole programme. See §6.

Falsified if — as expected — the limits are genuinely analytic, i.e. the degenerations exist only after passing to series with real parameters. The honest outcome is then a sub-poset: the part of the Askey scheme whose edges *are* container constructions, with the rest marked as outside.

Phase D — the q-layer

Deliverable. The q-analogue of Link 2: a container-level isomorphism whose fibre count is Proposition 3.1. By Corollary 3.2 the generators are already present, so the work is the relation.

Dependencies. Phase A (for the pattern to imitate); Phase 0.3; and *critically* Question 6.1 of `list-and-bag.tex` — does $\mathbf{Cont}_q$ have a usable K_0? — wherever a minus sign is needed.

Confidence. Medium-high (60%). Higher than Phase B because the target is a single relation rather than a general procedure, and because `hypergeometric.tex` §6 argues the q-side is the structurally honest one: there the obligatory factor $(q;q)_n$ is $|\mathrm{GL}_n(\mathbb{F}_q)|$ up to the torus, i.e. an automorphism group order by construction rather than by the accident that $(1)_n = |S_n|$.

Falsified if $K_0(\mathbf{Cont}_q)$ does not exist usefully. Then the q-layer is confined to statements with no subtraction in them, which excludes the commutator.

Phase E — Lean

Deliverable. Phase A formalised (this is already `lascoux/lean/`); the q-Weyl relation of Proposition 3.1 formalised, which is elementary; and whichever of B and D survive.

Dependencies. Whatever it formalises.

Confidence. High (80%) for what is in scope, because `lascoux/lean-plan.md` §2 has already inventoried Mathlib and the missing pieces are known: no Schur polynomials, no RSK, no Hall inner product.

Warning. Proposition 2.3 as stated needs the Hall inner product, which Mathlib does not have. Either prove $DU - UD = I$ combinatorially on Young diagrams — which is what step 1.8 does, and is the reason that step is phrased as a cardinality and not as an operator identity — or accept a substantial detour. Prefer the former. The operator phrasing is for this note, not for the kernel.

5 Dependency summary

phase	needs	confidence	status
0 orientation	—	n/a	not started
A Weyl at container level	—	85% / 50%	step 1.3 done
B WZ as container morphism	A, 0.2	35% / 75%	**open**
C Askey scheme as poset	A, B, 0.1	15%	**open**, see §6
D the q-layer	A, 0.3, K_0 question	60%	generators present
E Lean	the above	80%	`lascoux/lean/` live

6 What would kill the programme, and the one that probably will

Recorded separately from the phases, because a programme that cannot say how it fails is not a programme.

F1 Degeneration is analytic, not combinatorial. The edges of the Askey scheme are limits: $q \to 1$, a parameter $\to \infty$, a rescaling. Limits are not container operations; they are not operations on **Poly** at all, which is a rig category with no notion of convergence. *This is the expected failure, and it is Phase C.* The likely honest outcome is that the container reading covers the *vertices* of the scheme (each family, at integer parameters) and none of the *edges*, leaving a discrete set where a poset was wanted.

F2 No minus signs. **Poly** and $\mathbf{Cont}_q$ are rig categories. A commutator $DU - UD$ needs subtraction, so Link 2 at the container level works only because it is phrased as $|U \cap U| = \delta + |D \cap D|$ — all additions. Any step that genuinely needs the difference needs K_0, which is Question 6.1 of `list-and-bag.tex` and is open.

F3 Non-integral certificates. Phase B's general case, as above.

F4 The multiplicity space is a coincidence. If $\mathbb{C}[p_2, p_3, \dots]$ in Remark 2.4 has nothing to do with the higher parameters of the Askey families, then the bottom of the scheme is where the categorification lives and there is no ladder to climb.

Warning 6.1 (the shape of the honest outcome). Taking F1 and F4 together, the most likely true statement at the end of this programme is narrower than Thesis 2.1: *the Weyl algebra is*

categorified by containers; the Askey scheme is not. That would still be worth having. It would say that the organizing theme [1] lacks is the holonomic one, that the holonomic one has a combinatorial foundation, and that the scheme sitting on top of it is an analytic structure that the combinatorics does not see. Three sentences, all of them checkable, is a better outcome than an unfalsifiable programme.

7 Out of scope, with reasons

Asymptotics and orthogonality measures.
The analytic heart of [1]. Every technique there is estimation, and nothing in §2 bears on it. Not a gap — a different subject.

Painlevé and isomonodromy.
The natural continuation of the monodromy theme, which this note does not adopt. Deliberately left to one side; if Phase C fails as expected, monodromy is where to look next, because confluence of punctures is a genuinely structural operation in a way that parameter limits are not.

Multivariable and GKZ.
[4] organizes multivariable hypergeometric functions by toric geometry, and the polytope is plausibly a container-like object. This is the most interesting excluded item and should be revisited after Phase D, not before.

Macdonald polynomials.
Already scoped out of `lascoux/lean-plan.md`, for the reasons given there. The same reasons apply.

8 Open questions, collected

Question 8.1. Does the multiplicity space $\mathbb{C}[p_2, p_3, \dots]$ of Remark 2.4 correspond to the higher parameters of the Askey families? Concretely: is there an assignment of families in the scheme to subspaces or quotients of $\mathbb{C}[p_2, p_3, \dots]$ under which degeneration is a map of multiplicity spaces? If yes, F4 is answered and Phase C has a mechanism. If no, the programme stops at the Fock representation.

Question 8.2. Is there a container model of AW(3)? Equivalently: does Zhedanov's algebra have a presentation in which the relations are fibre-count identities with no subtraction, as $DU - UD = I$ does in the form $|U \cap U| = \delta + |D \cap D|$? This is the top-of-the-scheme question and it is the right one to ask before attempting Phase C, because a negative answer here closes Phase C immediately and cheaply.

Question 8.3. If Phase A's step 1.5 shows the local rule is an optic rather than an isomorphism — which thesis §4.2 half predicts — what does the *decategorification* of an optic look like? An isomorphism decategorifies to an equation. A lens decategorifies to what?

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